HEEKUN ROH / 노희건

hroh@mit.edu / heekunroh.github.io
Ph. D. Candidate at Massachusetts Institute of Technology, United States

EDUCATION

Massachusetts Institute of Technology (MIT) Sep. 2023 - Present
Ph. D. Candidate in Aeronautics and Astronautics
Advisor : Prof. Olivier L. de Weck (MIT Engineering Systems Laboratory)
Thesis : Complexity Quantification and Budgeting for Engineering Systems using Graph Based Abstraction
Keywords : Systems Engineering, Complexity, Design Process, Space Situational Awareness

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2017 - Feb. 2019
M. S. in Aerospace Engineering
Advisor : Prof. Min-Jea Tahk
Thesis : Impact time and angle control guidance for homing missiles using sequential convex programming
Keywords : Guidance and Control, Trajectory Optimization, Optimal Resource Allocation and Planning

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2013 - Feb. 2017
B. S. in Aerospace Engineering and Electrical Engineering
Summa Cum Laude, KAIST Presidential Fellow

PROFESSIONAL EXPERIENCE

Appointed Researcher, KAIST Dept. of Aerospace Eng. Dec. 2022 - Aug. 2023
Flight Dynamics and Control Lab (FDCL)
Keywords : Trajectory Optimization, Sequential Convex Optimization, In-House Optimizer Development

Associate Systems Engineer, Satrec Initiative Feb. 2019 - Oct. 2022
Keywords : Earth Observation Satellites, Satellite Attitude Determination and Control, Onboard Software Development and Testing, Mission Analysis, Verification and Validation, Commissioning

PUBLICATIONS

Journals

[1] H. Roh, and O. L. de Weck, "Propagation Complexity: A Metric for Graph Representation of Systems," (in preparation)

[2] H. Roh, J. J. Norheim, and O. L. de Weck, "Optimal Design Sequence for Multidisciplinary Analysis and Optimization," (in preparation)

[3] H. Roh, J. J. Norheim, and O. L. de Weck, "Spectral Clustering for Large Scale Design Structure Matrices," *ASME Journal of Mechanical Design - Special Issue on Extreme Design* (in revision)

[4] H. Roh, and O. L. de Weck, "Visibility Analysis of Uncatalogued Resident Space Objects with Constellation of In-Situ Optical Detectors," *AIAA Journal of Spacecraft and Rockets* (submitted)

[5] H. Roh, and O. L. de Weck, "Visibility Upper Bound of In-Situ Optics Constellation for Resident Space Objects," *AIAA Journal of Spacecraft and Rockets* (submitted)

[6] H. Roh, Y. J. Oh, M. J. Tahk, K. J. Kwon, and H. H. Kwon, "L1 Penalized Sequential Convex Programming for Fast Trajectory Optimization: With Application to Optimal Missile Guidance," *International Journal of Aeronautical and Space Sciences (IJASS)*, Vol. 21, pp. 493-503, Jun. 2020.

[7] H. Roh, Y. J. Oh, M. J. Tahk, and Y. R. Jung "Optimal Weapon-Target Assignment of Multiple Dissimilar Closed-In Weapon Systems Using Mixed Integer Linear Programming," *Journal of Korean Society for Aeronautical and Space Sciences*, Vol. 47, No. 11, pp.787-794, Nov. 2019.

[8] H. Roh, S. W. Shim, and M. J. Tahk, "Maneuver Algorithm for Bearings-Only Target Tracking with Acceleration and Field of View Constraints," *International Journal of Aeronautical and Space Sciences (IJASS)*, Vol. 19, No. 2, pp. 423-432, Jun. 2018.

[9] S. Y. Han, J. H. Bai, S. M. Hong, H. Roh, M. J. Tahk, J. S. Yun, S. H. Park, "Control law for agile turn of air-to- air missile during boost phase," *International Journal of Aeronautical and Space Sciences (IJASS)*, Vol.18, No.4, Dec. 2017

Conferences

[1] H. Roh, L. Etzenbach, A. Oltramare, J. Norheim, O. L. De Weck. "Factored Dependency Structure Matrix for Representation of Multi-Connection Systems," 27th International Dependency and Structure Modeling Conference (DSM 2025), Hoboken, NJ, USA, Sep. 2025.

[2] H. Roh, L. Etzenbach, A. Oltramare, J. Norheim, O. L. De Weck. "Size Constrained K-Means Clustering for Controlled Design Structure Matrix Partitioning," 2025 IEEE International Systems Conference (SysCon), Montreal, Canada, Apr. 2025.

[3] L. Etzenbach, H. Roh, A. Oltramare, J. Norheim, O. L. De Weck. "The Road Ahead: Present and Future Trends in Global Civil Aircraft Regulations," 2025 IEEE International systems Conference (SysCon), Montreal, Canada, Apr. 2025.

[4] L. Etzenbach, H. Roh, A. Oltramare, J. Norheim, O. L. De Weck. "Design-Driven Framework for the Complexity Quantification of Mechanical Components," 2025 IEEE International systems Conference (SysCon), Montreal, Canada, Apr. 2025.

[5] H. Roh, and O. L. De Weck, "Global Visibility Assessment Method of In-Situ Optics Constellation for Uncatalogued Resident Space Objects," *AIAA Scitech Forum 2025*, Orlando, Florida, USA, Jan. 2025.

[6] M. J. Tahk, J. C. Park, H. Roh, C. H. Lee., "Suboptimal guidance based on pursuit and impact angle control for long-range air-to-air missiles," 14th International Conference on Mechanical and Aerospace Engineering (ICMAE), Porto, Portugal, Jul. 2023

[7] J. C. Park, H. Roh, C. H. Lee, M. J. Tahk, H. S. Choi, S. Y. Kim. "Nonlinear Three-Loop Autopilot Design for Hypersonic Vehicles Considering Aerodynamic-Propulsion Dynamics Couplings," 14th International Conference on Mechanical and Aerospace Engineering (ICMAE), Porto, Portugal, Jul. 2023

[8] H. Roh, T. Kim, Y. Moon, H. Woo, C. H. Lee, "Optimal Time Terrestrial Mapping Scheduling for Spaceborne Push-Broom Imagers," Aerospace Europe Conference 2023 (EUCASS-CEAS 2023), Lausanne, Switzerland, Jul. 2023.

[9] Y. J. Oh, H. Roh, and M. J. Tahk, "Fast Trajectory Optimization using Sequential Convex Programming with No-Fly Zone Constraints," 21st IFAC Symposium on Automatic Control in Aerospace (ACA), Cranfield, United Kingdom, Aug. 2019.

[10] H. Roh, Y. J. Oh, M. J. Tahk, and C. H. Lee, "Fast Trajectory Optimization Using Sequential Convex Method for Guided Missiles," *The 5th CEAS Conference on Guidance, Navigation and Control (EuroGNC)*, Milano, Italy, Apr. 2019.

[11] H. Roh, M. J. Tahk, K. B. Kim, and H. H. Kwon, "Trajectory Optimization for Missile Impact Time Control Problem Using L1 Penalty Method and Sequential Convex Programming," *The Korean Society for Aeronautical and Space Sciences(KSAS): 2018 Fall Conference*, Jeju, Korea, Nov. 2018.

[12] H. Roh, and M. J. Tahk, "Comparison Study on Bearings-Only Target Tracking Filters," *The Society for Aerospace System Engineering (SASE): 2018 Fall Conference*, Gyeongju, Korea, Nov. 2018.

[13] Y. J. Oh, H. Roh, M. J. Tahk, "A Lattice Path Following Algorithm for Guided Missiles," *2018 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Chengdu, China, Oct. 2018.

[14] H. Roh, S. M. Hong, M. J. Tahk, K. B. Kim, K. J. Kwon, and H. H. Kwon, "Optimal Impact Time Control Guidance Using Convex Optimization ," *Korean Institute of Military Science and Technology (KIMST): 2018 General Conference*, Jeju, Korea, Jun. 2018.

[15] H. Roh, and M. J. Tahk, "Optimization of Closed-In Weapon System Target Assignment Using Mixed Integer Linear Programming," *The Korean Society for Aeronautical and Space Sciences (KSAS): 2018 Spring Conference*, Goseong, Korea, Apr. 2018.

[16] H. Roh, M. H. Cho and M. J. Tahk, "Trajectory Optimization Using Cramer-Rao Lower Bound for Bearings-Only Target Tracking," *AIAA Scitech Forum 2018*, Kissimmee, Florida, USA, Jan. 2018.

[17] H. Roh, B. Y. Lee, and M. J. Tahk, "Maneuver Generation for Moving Obstacle Avoidance Using Scoring Function Matrix," *The Korean Society for Aeronautical and Space Sciences (KSAS): 2017 Fall Conference*, Jeju, Korea, pp. 561-562, Nov. 2017.

[18] H. Roh, B. Y. Lee, and M. J. Tahk, "Automatic Maneuver Generation for Suppression of Enemy Air Defense Using Scoring Function Matrix," *Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, pp.1087-1093, Seoul, Korea, Oct. 2017.

[19] H. Roh, J. M. Park, and M. J. Tahk, "Modeling and Formulation for Return-to-Launch-Site Trajectory Optimization of Reusable Launch Vehicle," *The 17th Symposium on Space Launch Vehicle Technology*, Goheung, Korea, Aug. 2017

[20] J. M. Park, S. M. Hong, H. Roh, M. J. Tahk, Y. Y. Kim, J. S. Yun, "Optimal control of roll-pitch seeker with singularity avoidance," *The 26th Mediterranean Conference on Control and Automation (MED)*, Zadar, Croatia, Jun. 2018.

[21] J. M. Park, H. Roh, M. J. Tahk, "Co-evolutionary Method For Dynamic Weapon-Target Assignment," *Advances in Control and Optimization of Dynamic Systems(ACODS)*, Hyderabad, India, Feb. 2018.

[22] S. Y. Han, J. H. Bai, H. Roh, S. M. Hong, M. J. Tahk, J. S. Yun, S. H. Park, "Three-Dimensional Velocity Maximizing Agile Turn of Air-to-Air Missile with Collision Triangle Constraint," *25th Mediterranean Conference on Control and Automation (MED)*, Valletta, Malta, Jul. 2017.

[23] K. Kim, H. Kim, H. Roh, and H. L. Choi, "Flying BioLab : A CanSat platform for sampling and monitoring air bacteria in bio-hazardous area" *The Korean Society for Aeronautical and Space Sciences (KSAS): 2014 Fall Conference*, Jeju, Korea, Nov. 2014.

FUNDED RESEARCH PROJECTS

(MIT Strategic Engineering Research Group)

Optimizing Product Lifecycle through Systems Engineering Maturity Integration (GE Vernova) Sep. 2026 - Aug. 2027

Developing digitally enabled framework for assessing system maturity and improving lifecycle performance, in accordance with GE Vernova's Engineering 2030 vision

Complexity Quantification and Budgeting in System Design Jan. 2024 - Dec. 2026
(Airbus)
Developing method for quantifying complexity of engineering systems, and assessing its potential for engineering management and budgeting

Space-WATCH Marketplace Architecture Design Jan. 2024 - Aug. 2025
(DARPA / MIT Lincoln Laboratory)
Analyzing the capability of in-situ sensor-based space situational awareness, and exploring potential collaboration with ground-based, high-performance detectors

PUBLIC CONTRIBUTION

Invited Lectures

- CentraleSupélec (FRA), Measuring System Complexity (2026)
- Tecnológico de Monterrey (MEX), A Brief Introduction to Space Systems Engineering (2026)
- Boryung Pharmaceuticals (KOR) visit to MIT Media Lab, Complexity Management for System Design and Development (2024, 2025)
- Korea Advanced Institute of Science and Technology (KOR), Complexity Management for System Design and Development (2025)
- Sejong University (KOR), Complexity Management for System Design and Development (2025)
- Gyeongsang National University (KOR) visit to MIT ESL, Complexity Management for System Design and Development (2024)

Professional Memberships

- The International Council on Systems Engineering (INCOSE)
- The American Institute of Aeronautics and Astronautics (AIAA)
- The Korean Society for Aeronautical and Space Sciences (KSAS)

Reviewer Activities

- IEEE Transactions on Aerospace and Electronic Systems (2022, 2024, 2026)
- AIAA Journal of Spacecraft and Rockets (2024)
- International Journal of Aeronautical and Space Sciences (2025, 2026)
- (Conference) Complex Adaptive Systems (2026)

Open Source Contributions

- C89 Implementation of the MINRES Algorithm (2023)
  https://web.stanford.edu/group/SOL/software/minres/

Community Services

- MIT AeroAstro dREFS: departmental Resources for Easing Friction and Stress (2024-)

HONORS & AWARDS

Graduation with Highest Honors, Summa Cum Laude Feb. 2017

KAIST Presidential Fellowship Mar. 2013 - Feb. 2017
Selected 20 students in the institute who show academic excellence and leadership

Boeing Scholarship Mar. 2014 - Dec. 2016
Selected students in the Department of Aerospace Engineering, KAIST

Dean's List, College of Engineering Mar. 2015

The World Embedded Software Contest 2014, 1st Place Dec. 2014
in 'Medical Services' section
Organized by Ministry of Trade, Industry and Energy

Samsung SDS Software Club Championship 2014, 1st Place Nov. 2014
Organized by Samsung SDS

Cansat Competition Korea 2014, 2nd Place Aug. 2014
Organized by KAIST Satellite Technology Research Center (SATREC)

Cansat Competition Korea 2012, 1st Place Aug. 2012
Organized by KAIST Satellite Technology Research Center (SATREC)

LANGUAGE PROFICIENCY

Korean
Native

English
Full Professional Proficiency
TOEFL 110 - R30/L30/S25/W25 (2022)
GRE V165/Q170/AW5.0 (2022)

French
Advanced Working Proficiency
DELF B2 (2016)

TECHNICAL SKILLS

Programming Languages	Core: C, Python, MATLAB (Daily production use) Developed flight control software for in-orbit EO Satellites in C
Optimization Softwares	Core: MOSEK, GPOPS II, Gurobi Developed in-house QP, SOCP interior point solvers in C, MATLAB
Miscellaneous	Agentic AI Workflow (Goose / Pi), Atlassian SWs, GMAT / STK

CERTIFICATES

AeroAstro Teaching in Engineering Certificate (AATEC)
Suite of courses and hands-on active teaching sessions for 8 months (in progress)

MIT Professional Development Certificate in Grant Writing
12+ hours of training sessions targeted for effective grant writing